

Boston Housing Data - Full Exploratory Visualization

Set Working Directory

```
setwd("/Users/Ian/Documents/Code_Projects/boston-house-price-modeling/visualization")
getwd()
```

```
## [1] "/Users/Ian/Documents/Code_Projects/boston-house-price-modeling/visualization"
```

Load Libraries

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(patchwork)
```

Load Data

```
data <- read.csv("../data/boston_cleaned.csv")
str(data)
```

```
## 'data.frame':   506 obs. of  13 variables:
##  $ CRIM      : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
##  $ ZN        : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
```

```
## $ INDUS : num 2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ CHAS : int 0 0 0 0 0 0 0 0 0 0 ...
## $ NOX : num 0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ RM : num 6.58 6.42 7.18 7 7.15 ...
## $ AGE : num 65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ DIS : num 4.09 4.97 4.97 6.06 6.06 ...
## $ RAD : int 1 2 2 3 3 3 5 5 5 5 ...
## $ TAX : int 296 242 242 222 222 222 311 311 311 311 ...
## $ PTRATIO: num 15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ LSTAT : num 4.98 9.14 4.03 2.94 5.33 ...
## $ MEDV : num 24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

```
summary(data)
```

```
##          CRIM          ZN          INDUS          CHAS
## Min.   : 0.00632   Min.   : 0.00   Min.   : 0.46   Min.   :0.00000
## 1st Qu.: 0.08205   1st Qu.: 0.00   1st Qu.: 5.19   1st Qu.:0.00000
## Median : 0.25651   Median : 0.00   Median : 9.69   Median :0.00000
## Mean   : 3.61352   Mean   : 11.36   Mean   :11.14   Mean   :0.06917
## 3rd Qu.: 3.67708   3rd Qu.: 12.50   3rd Qu.:18.10   3rd Qu.:0.00000
## Max.   :88.97620   Max.   :100.00   Max.   :27.74   Max.   :1.00000
##          NOX          RM          AGE          DIS
## Min.   :0.3850   Min.   :3.561   Min.   : 2.90   Min.   : 1.130
## 1st Qu.:0.4490   1st Qu.:5.886   1st Qu.: 45.02   1st Qu.: 2.100
## Median :0.5380   Median :6.208   Median : 77.50   Median : 3.207
## Mean   :0.5547   Mean   :6.285   Mean   : 68.57   Mean   : 3.795
## 3rd Qu.:0.6240   3rd Qu.:6.623   3rd Qu.: 94.08   3rd Qu.: 5.188
## Max.   :0.8710   Max.   :8.780   Max.   :100.00   Max.   :12.127
##          RAD          TAX          PTRATIO          LSTAT
## Min.   : 1.000   Min.   :187.0   Min.   :12.60   Min.   : 1.73
## 1st Qu.: 4.000   1st Qu.:279.0   1st Qu.:17.40   1st Qu.: 6.95
## Median : 5.000   Median :330.0   Median :19.05   Median :11.36
## Mean   : 9.549   Mean   :408.2   Mean   :18.46   Mean   :12.65
## 3rd Qu.:24.000   3rd Qu.:666.0   3rd Qu.:20.20   3rd Qu.:16.95
## Max.   :24.000   Max.   :711.0   Max.   :22.00   Max.   :37.97
##          MEDV
## Min.   : 5.00
## 1st Qu.:17.02
## Median :21.20
## Mean   :22.53
## 3rd Qu.:25.00
## Max.   :50.00
```

Attribute Information

Input features in order:

- 1) CRIM: per capita crime rate by town
- 2) ZN: proportion of residential land zoned for lots over 25,000 sq.ft.
- 3) INDUS: proportion of non-retail business acres per town
- 4) CHAS: Charles River dummy variable (1 if tract bounds river; 0 otherwise)
- 5) NOX: nitric oxides concentration (parts per 10 million) *[parts/10M]*

- 6) RM: average number of rooms per dwelling
- 7) AGE: proportion of owner-occupied units built prior to 1940
- 8) DIS: weighted distances to five Boston employment centres
- 9) RAD: index of accessibility to radial highways
- 10) TAX: full-value property-tax rate per \$10,000 [$\$/10k$]
- 11) PTRATIO: pupil-teacher ratio by town
- 12) LSTAT: % lower status of the population

Output variable:

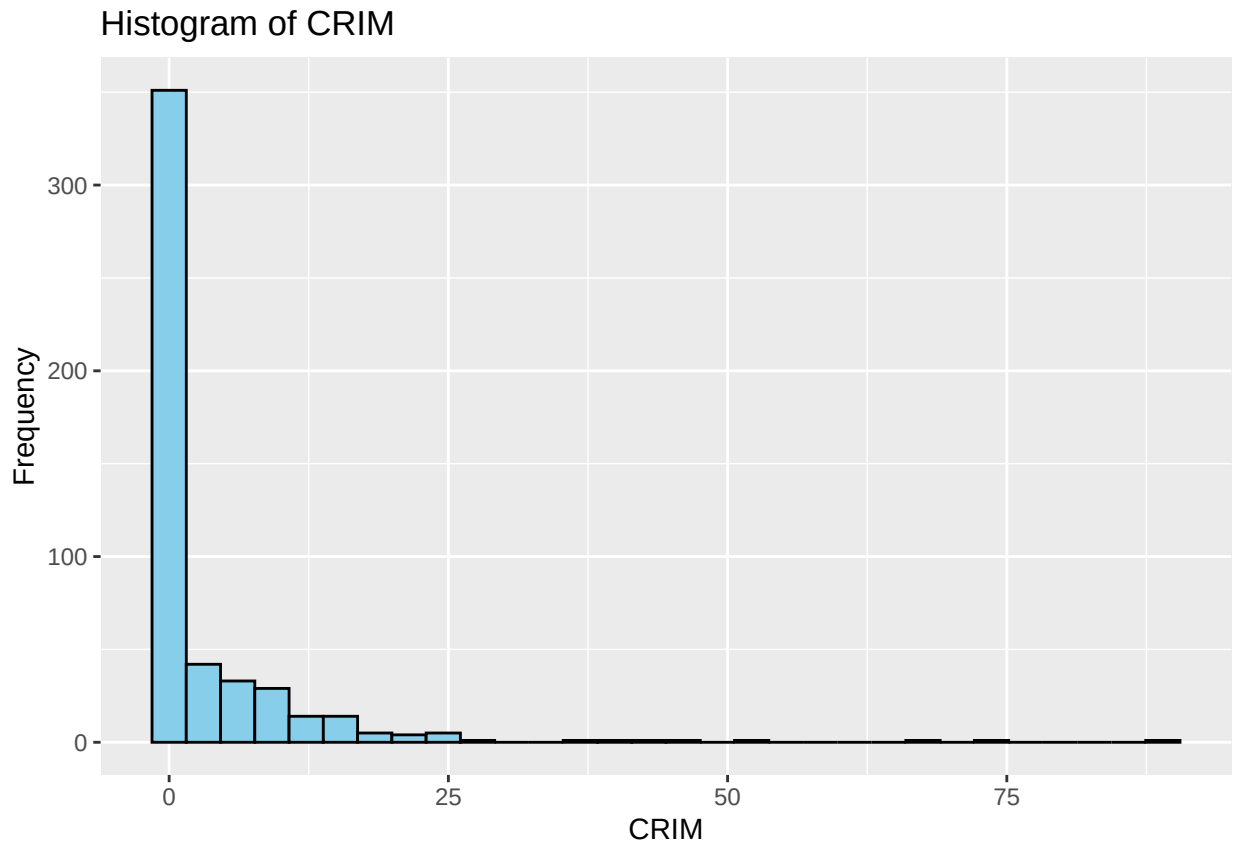
- 1) MEDV: Median value of owner-occupied homes in \$1000's [$k\$$]

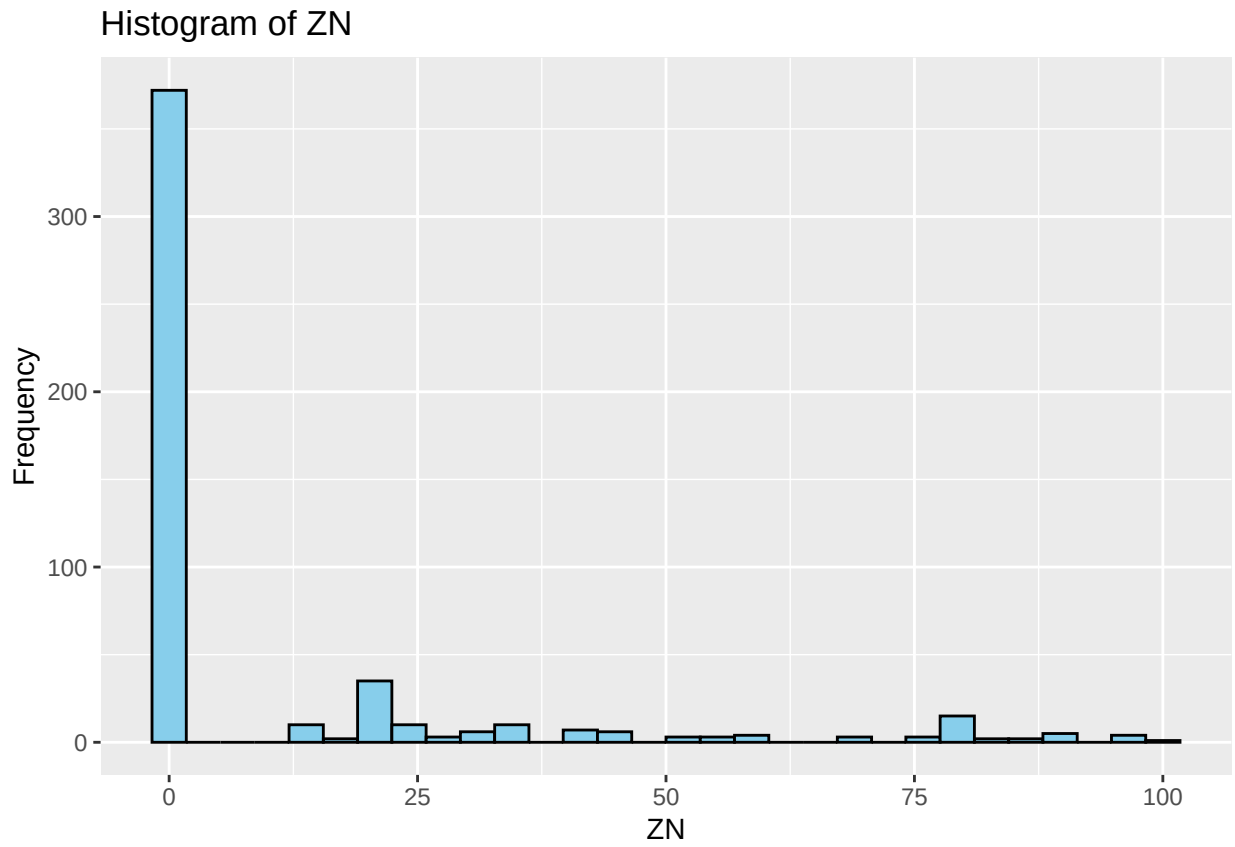
Part 1: Distribution of All Predictive Variables

```
predictor_vars <- names(data)[names(data) != "MEDV"]

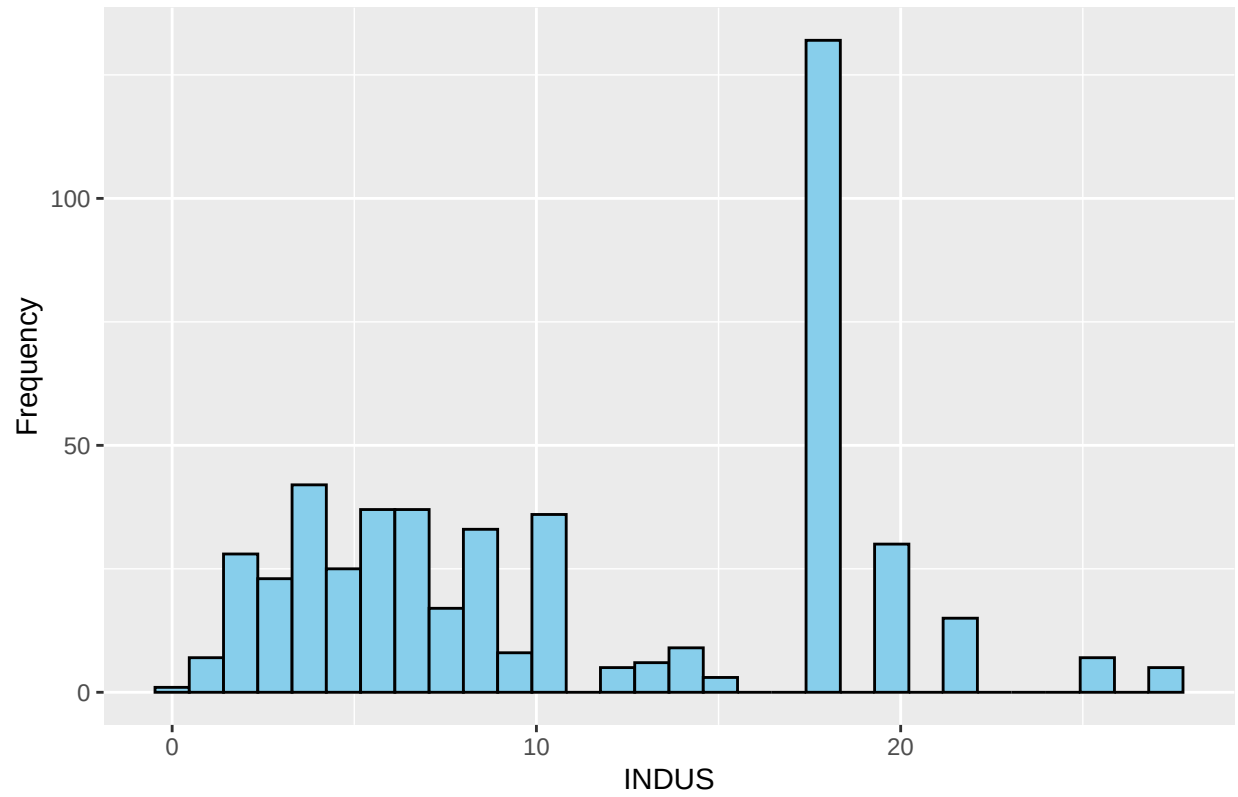
# Generate histograms for each predictor
for (var in predictor_vars) {
  print(
    ggplot(data, aes_string(x = var)) +
      geom_histogram(fill = "skyblue", color = "black", bins = 30) +
      labs(title = paste("Histogram of", var), x = var, y = "Frequency")
  )
}
```

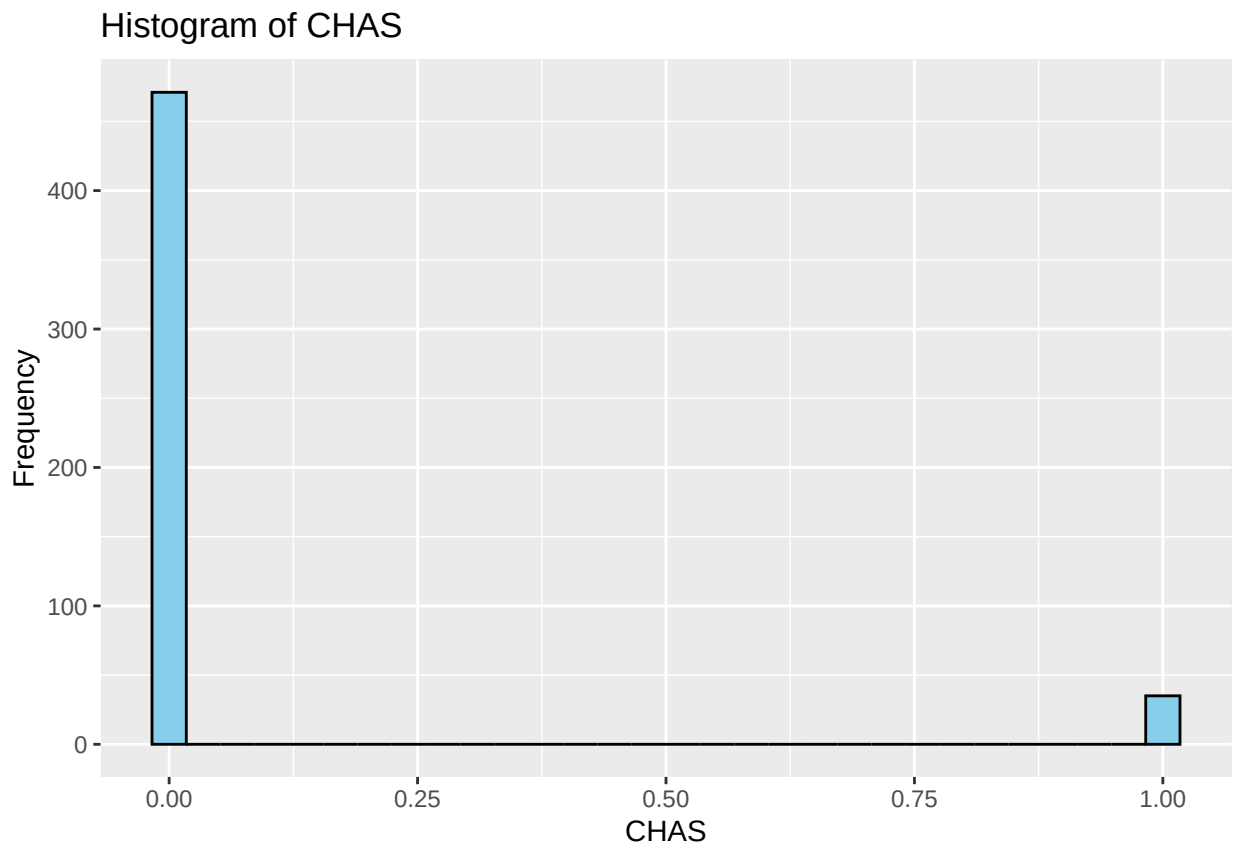
```
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

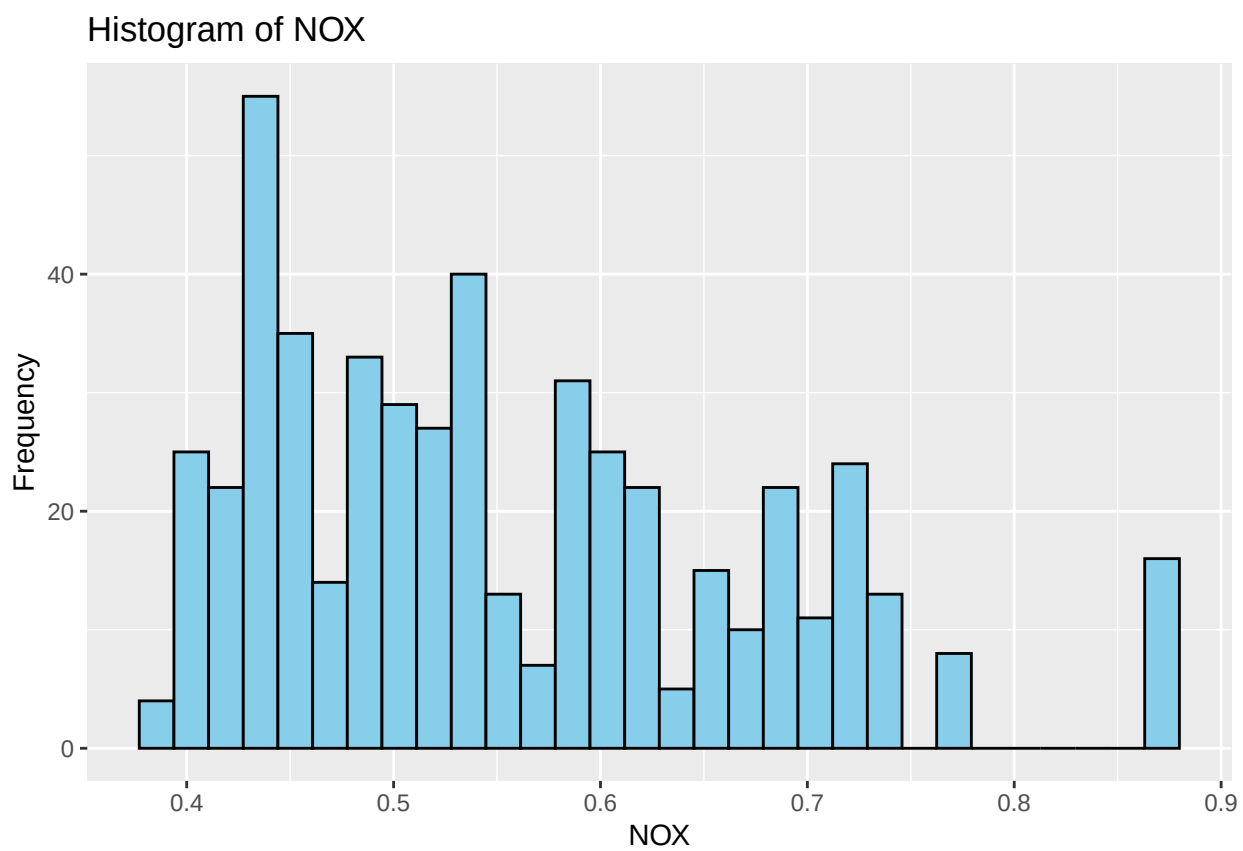


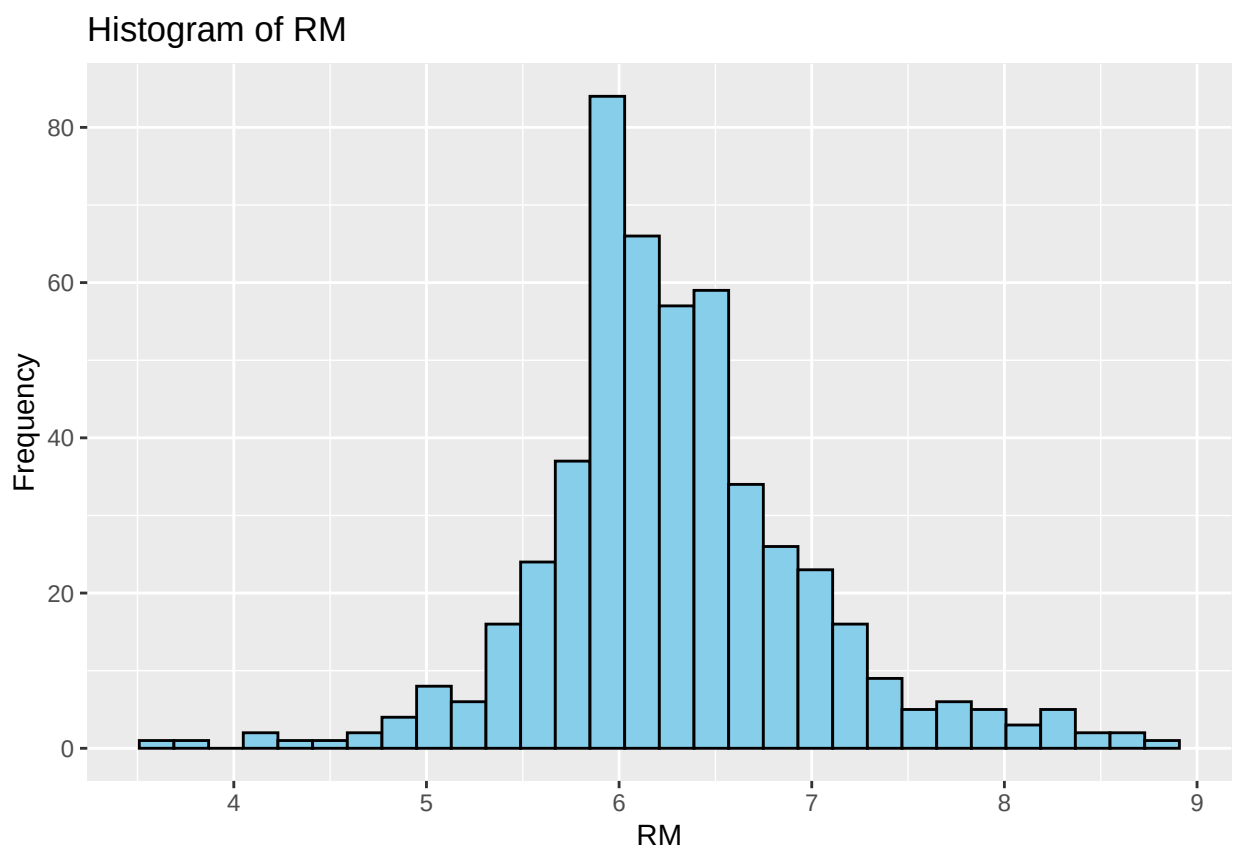


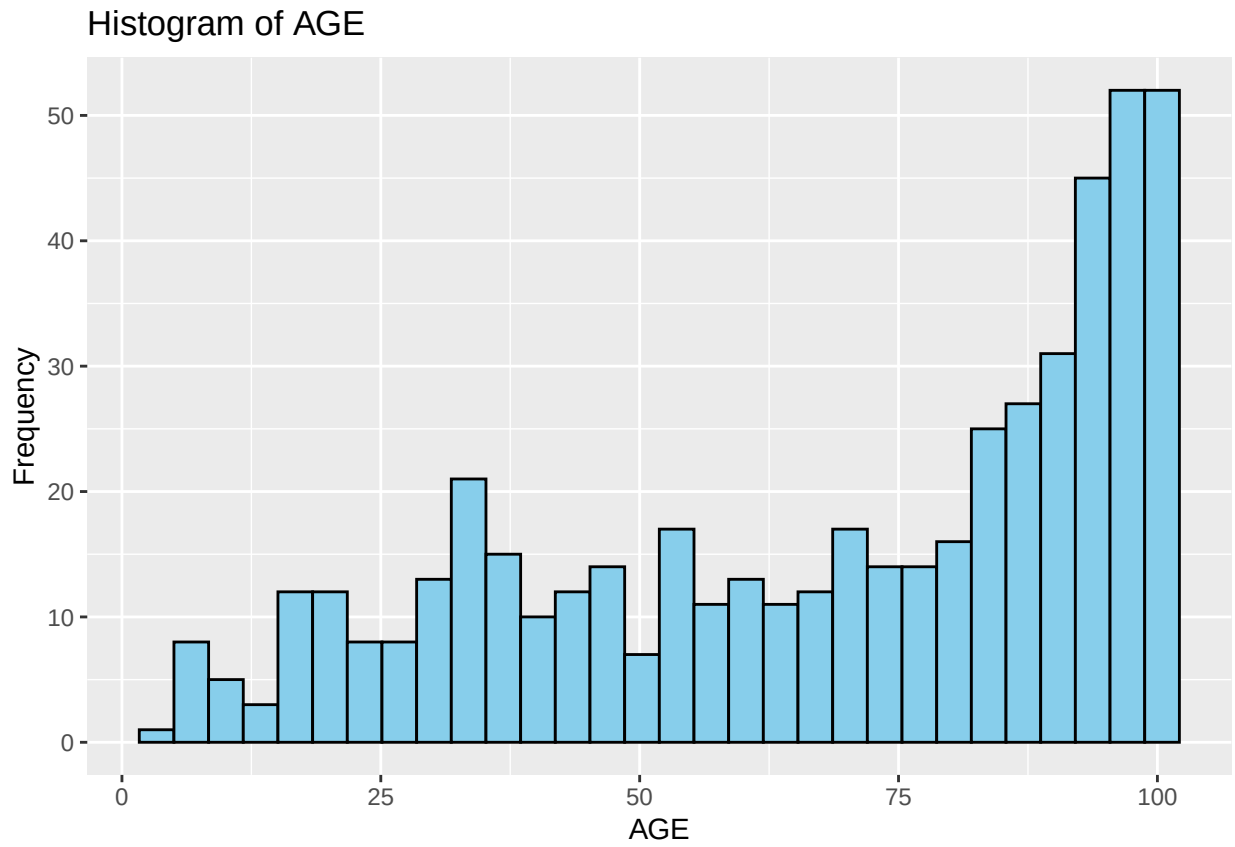
Histogram of INDUS



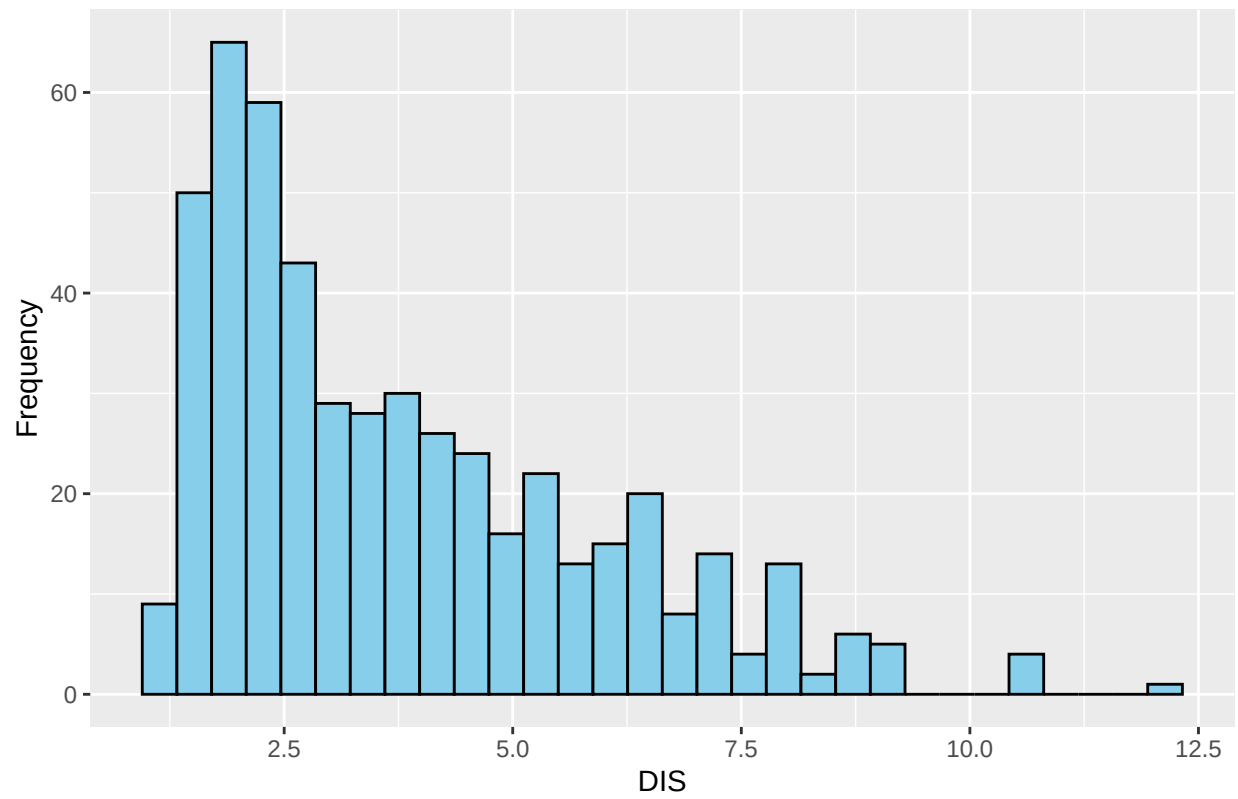


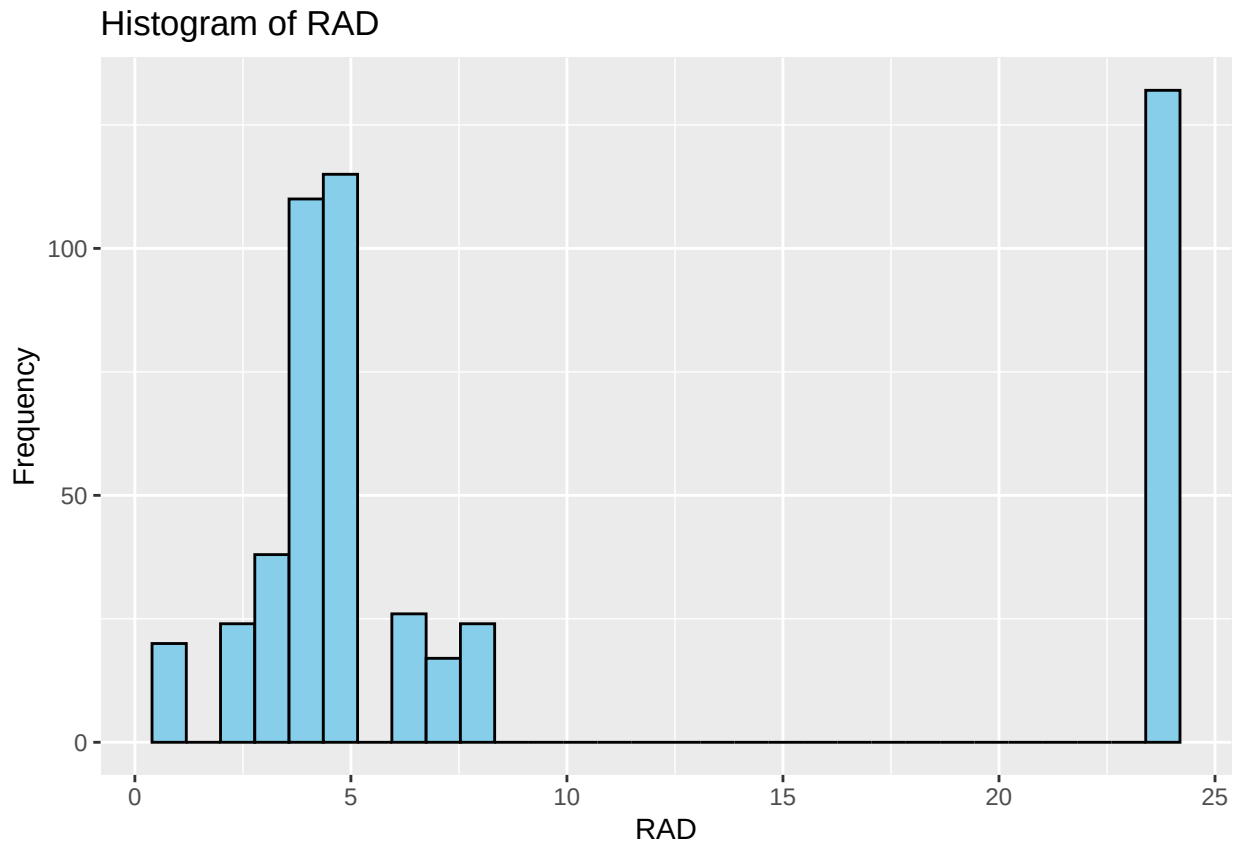




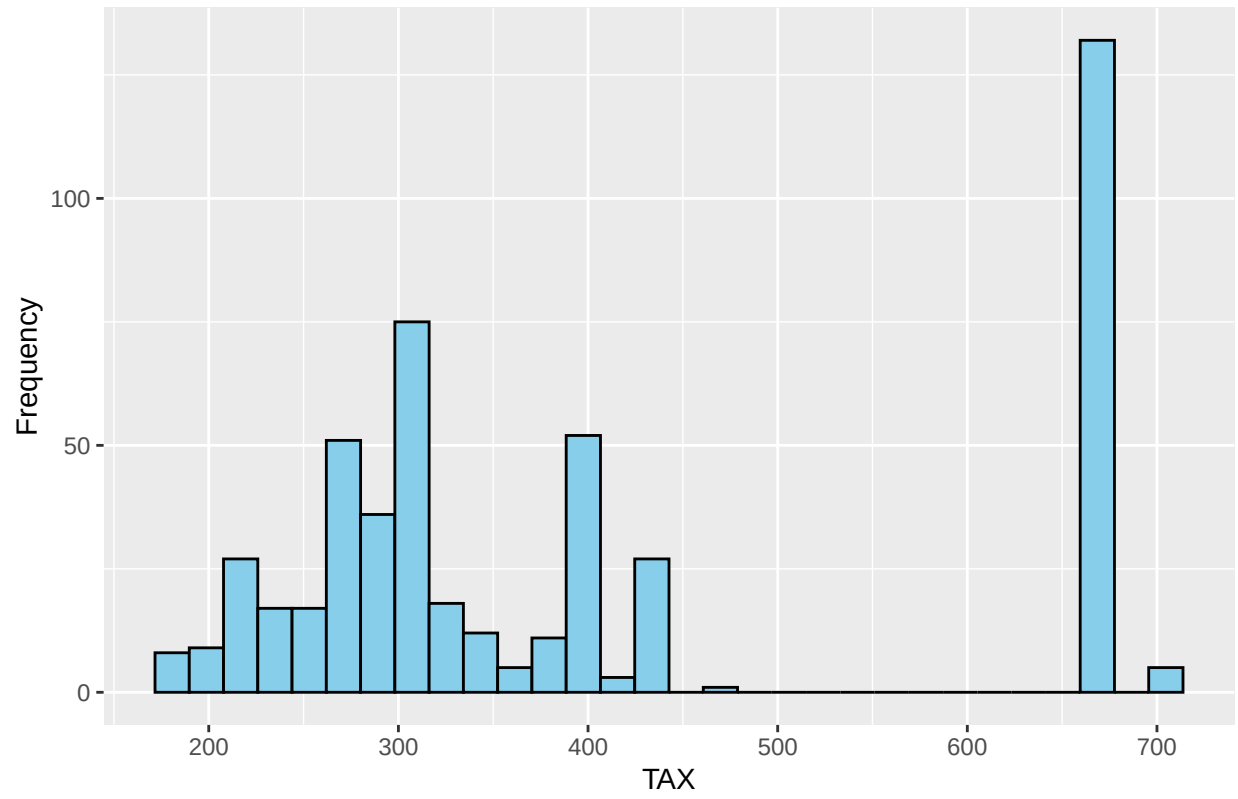


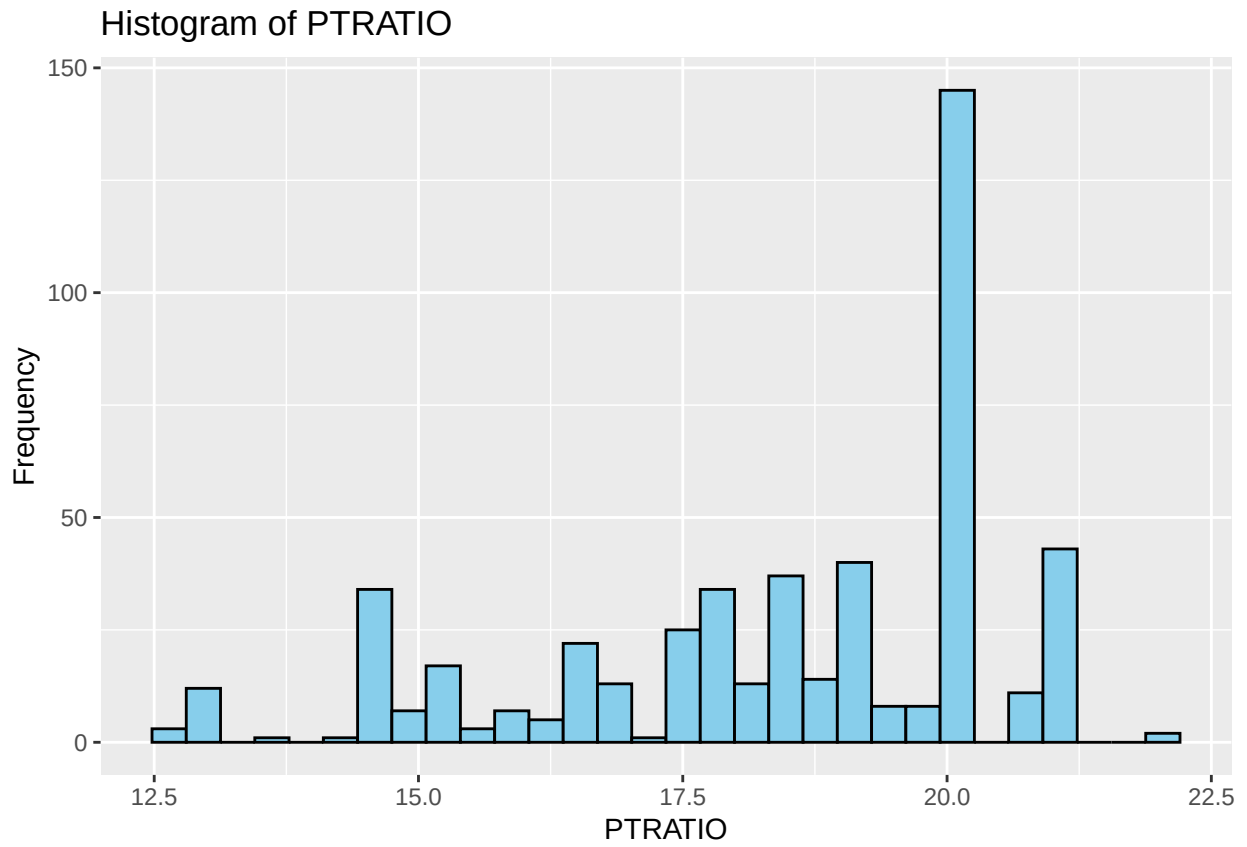
Histogram of DIS

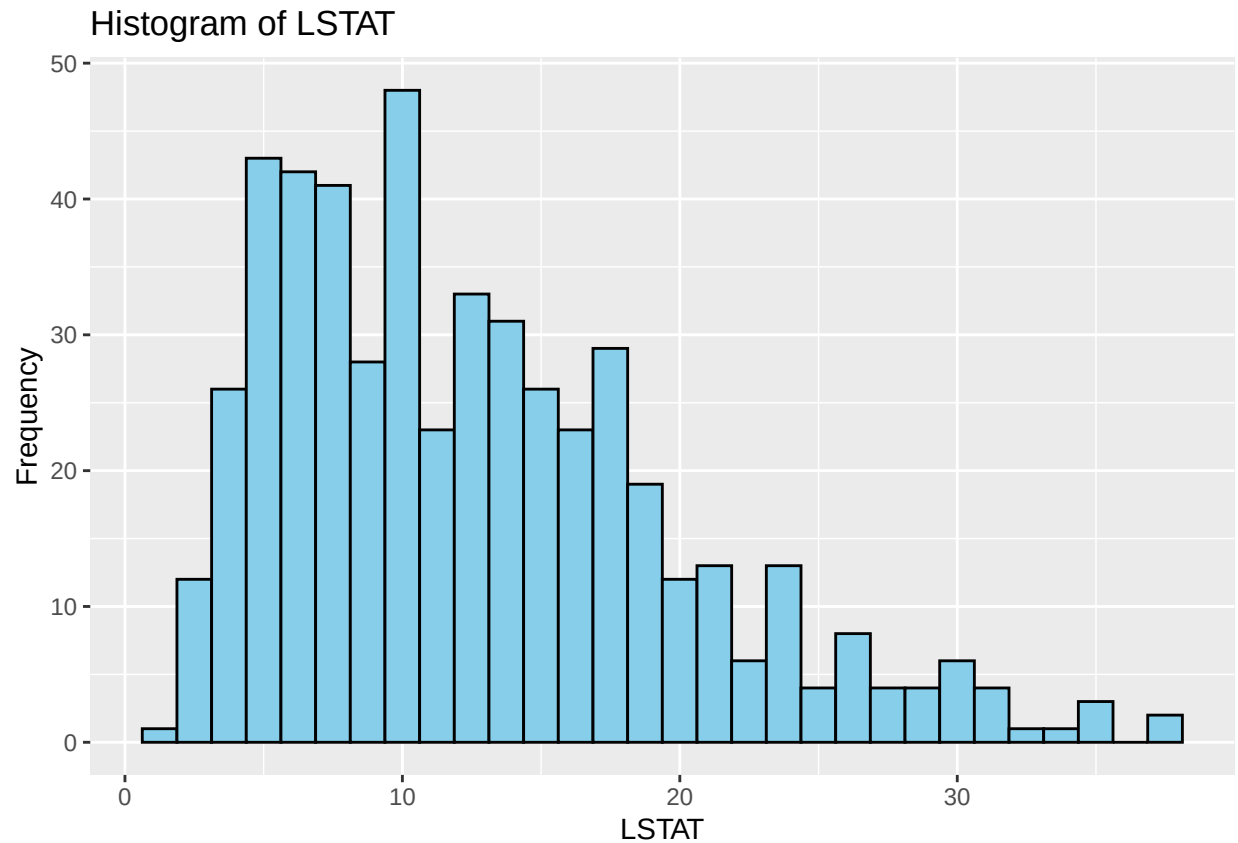




Histogram of TAX

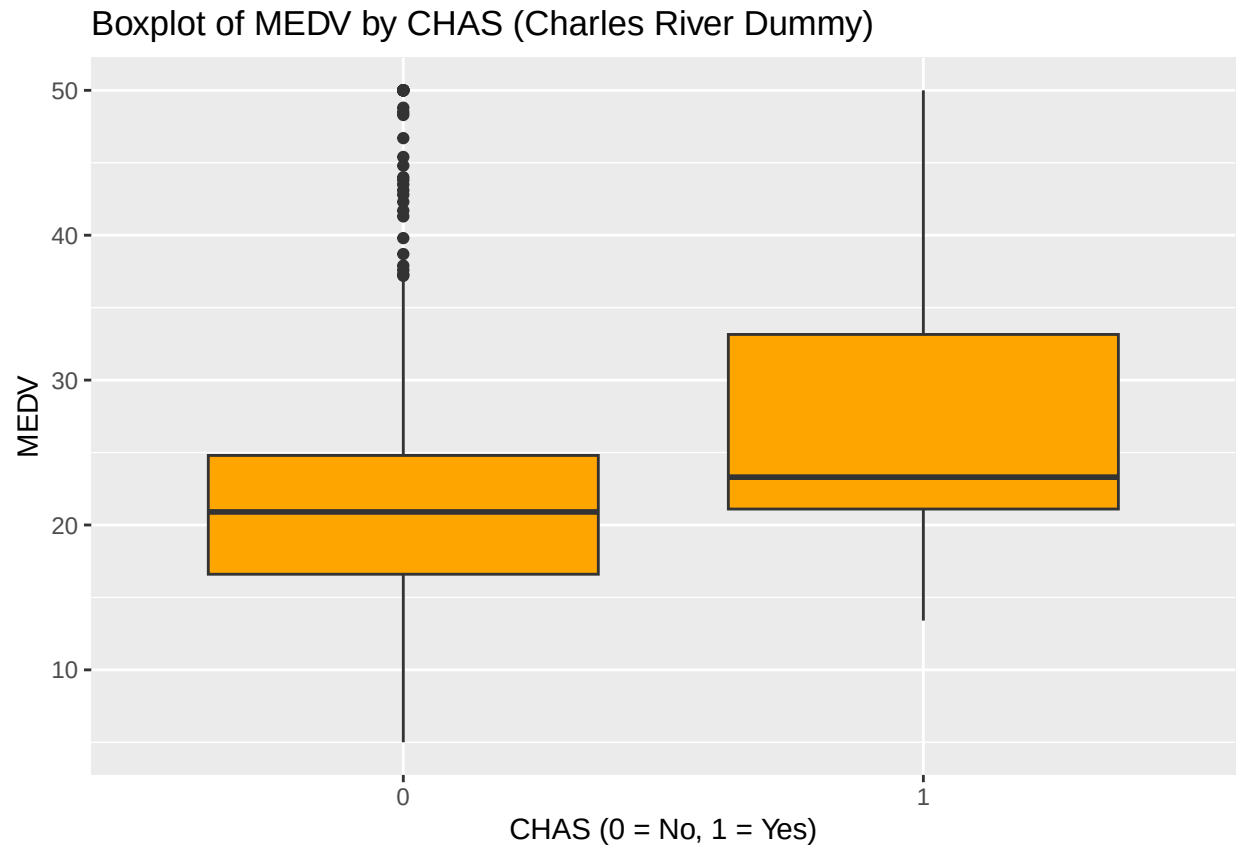






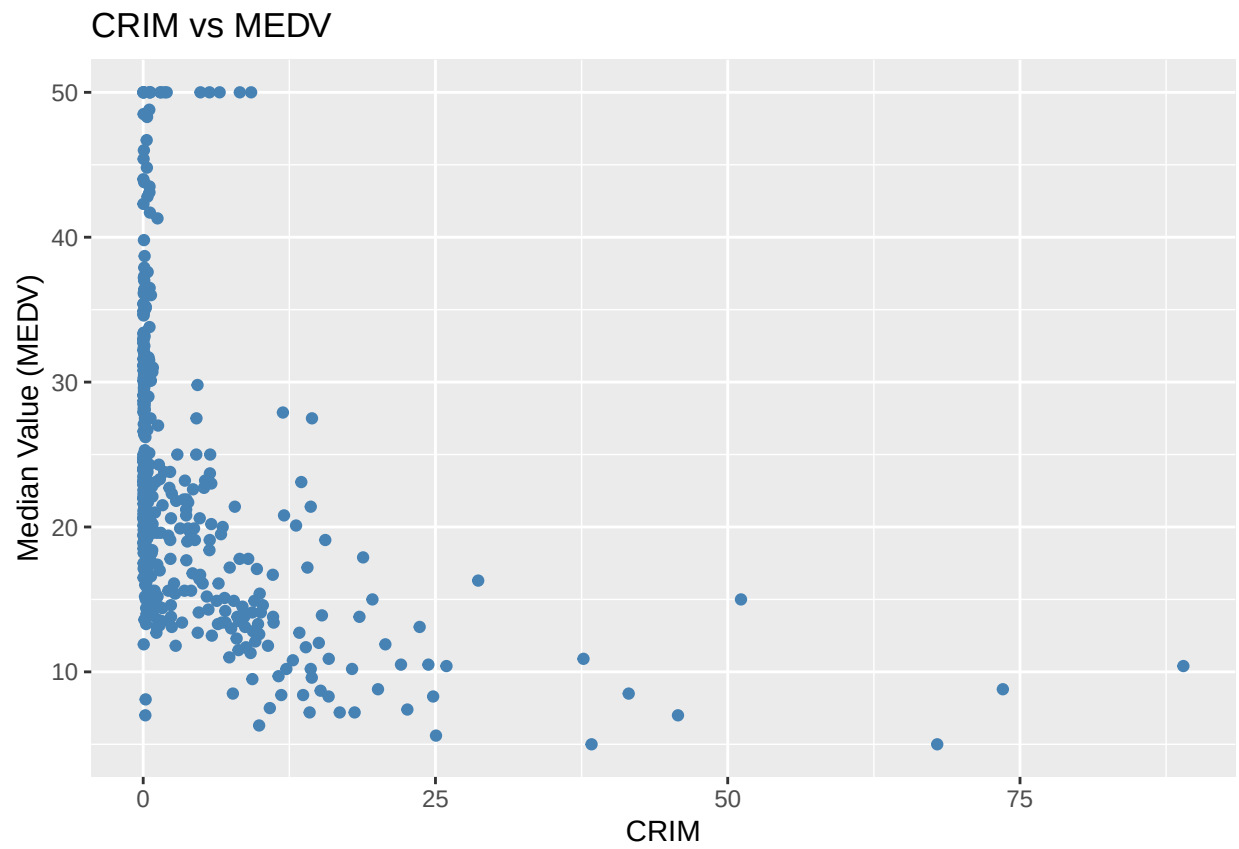
Part 2: Relationship Between Predictors and MEDV

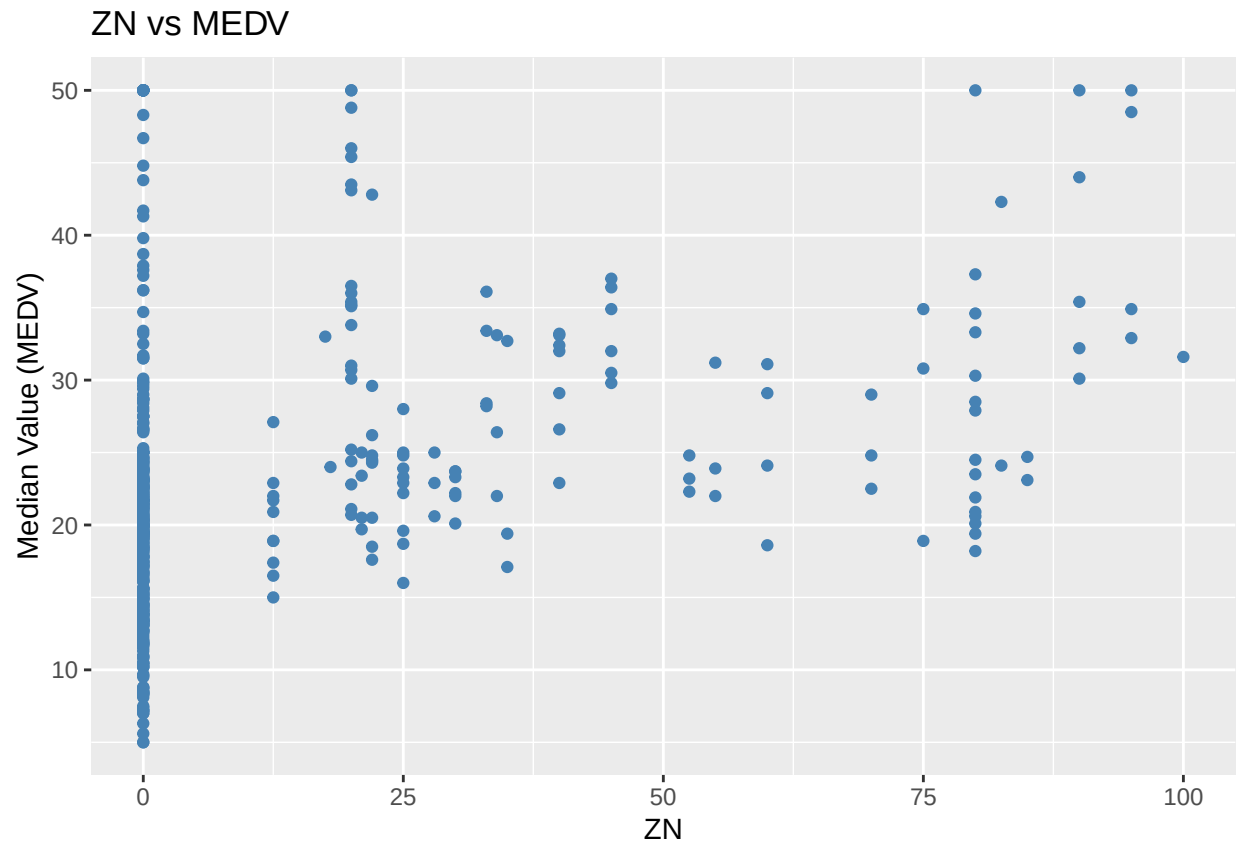
```
# Special case for CHAS (categorical)
ggplot(data, aes(x = factor(CHAS), y = MEDV)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot of MEDV by CHAS (Charles River Dummy)",
       x = "CHAS (0 = No, 1 = Yes)", y = "MEDV")
```

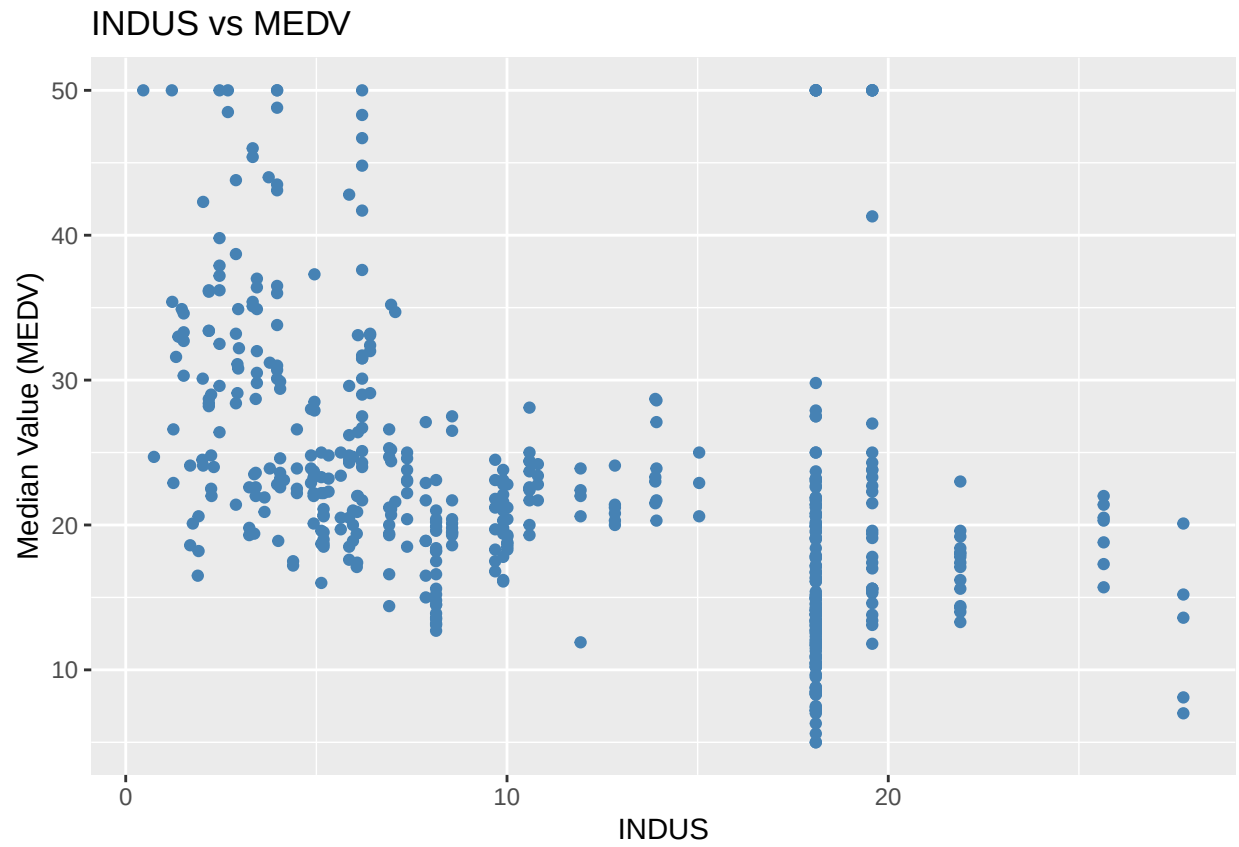


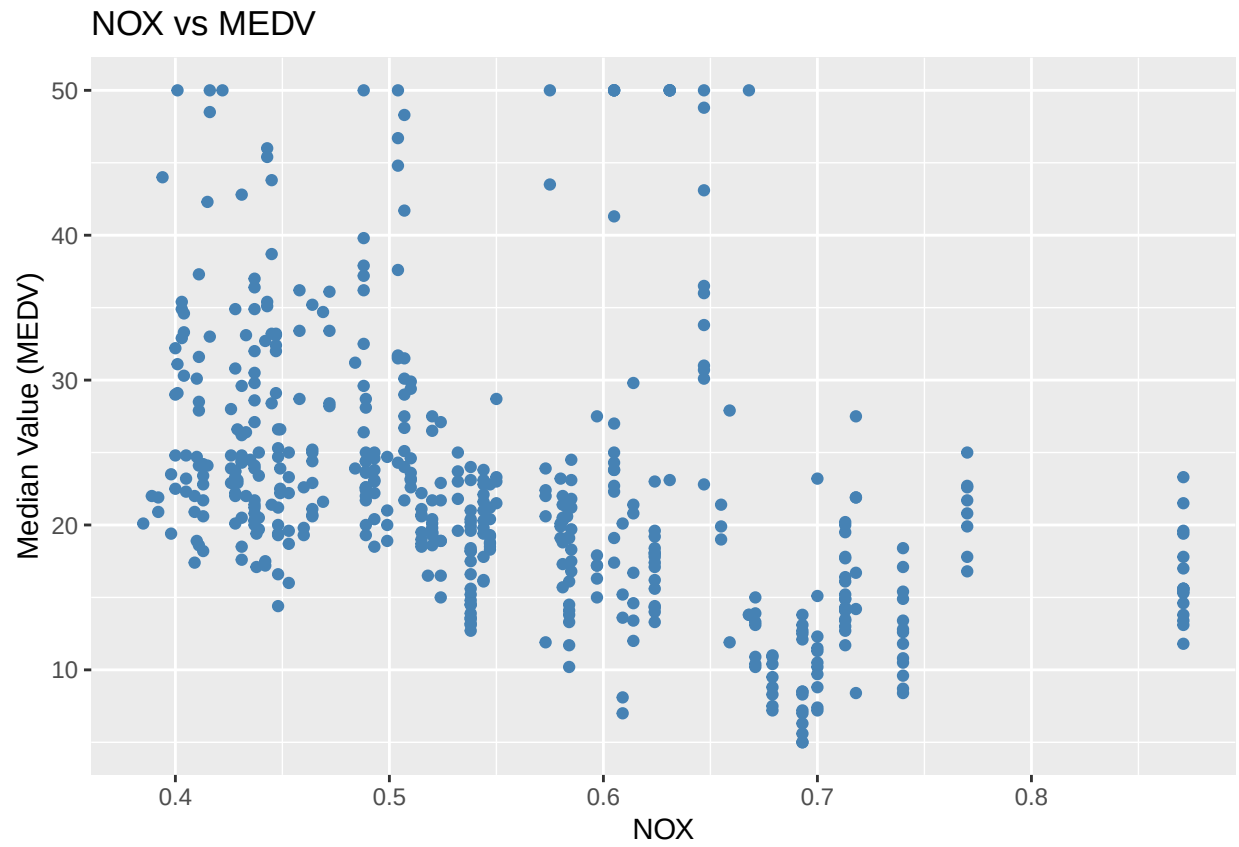
```
# Scatter plots for all other numeric predictors vs MEDV
numeric_vars <- setdiff(predictor_vars, "CHAS")

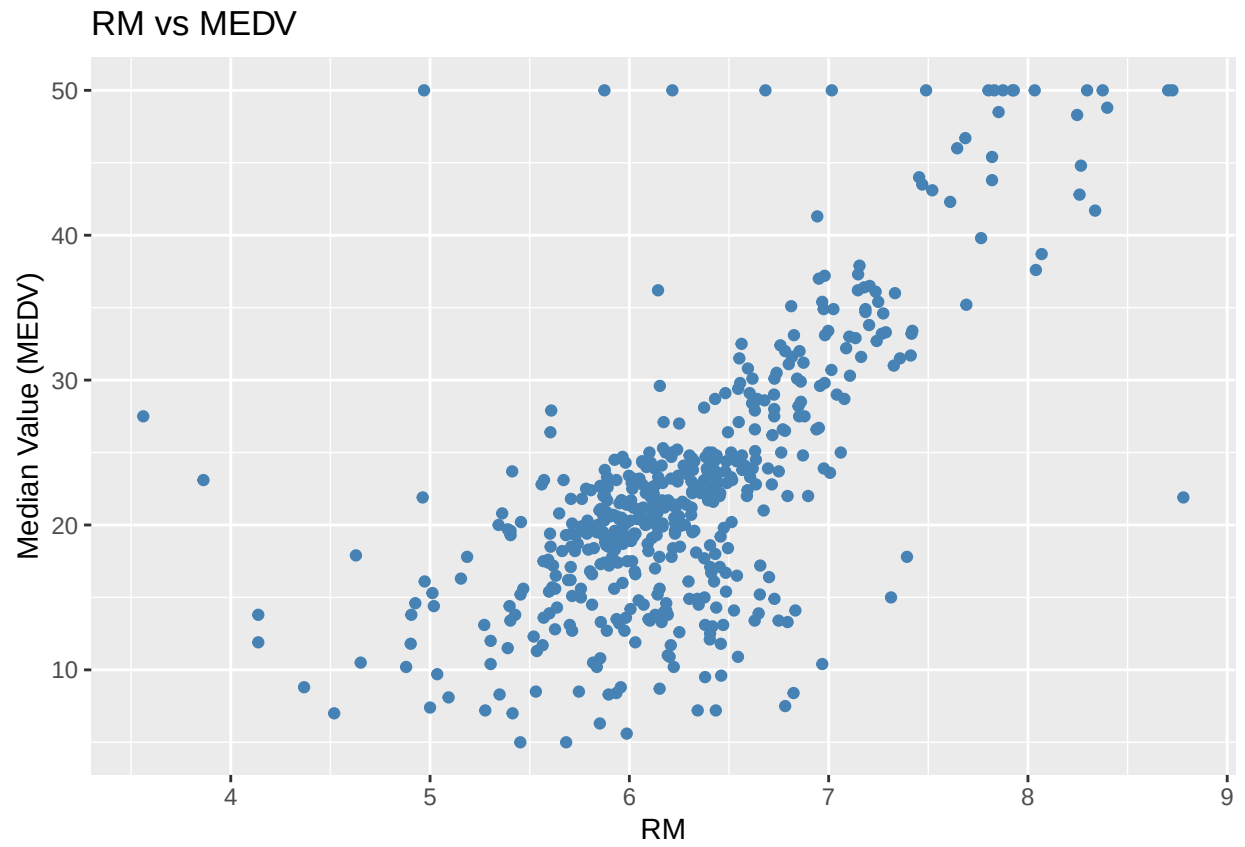
for (var in numeric_vars) {
  print(
    ggplot(data, aes_string(x = var, y = "MEDV")) +
      geom_point(color = "steelblue") +
      labs(title = paste(var, "vs MEDV"),
           x = var, y = "Median Value (MEDV)")
  )
}
```



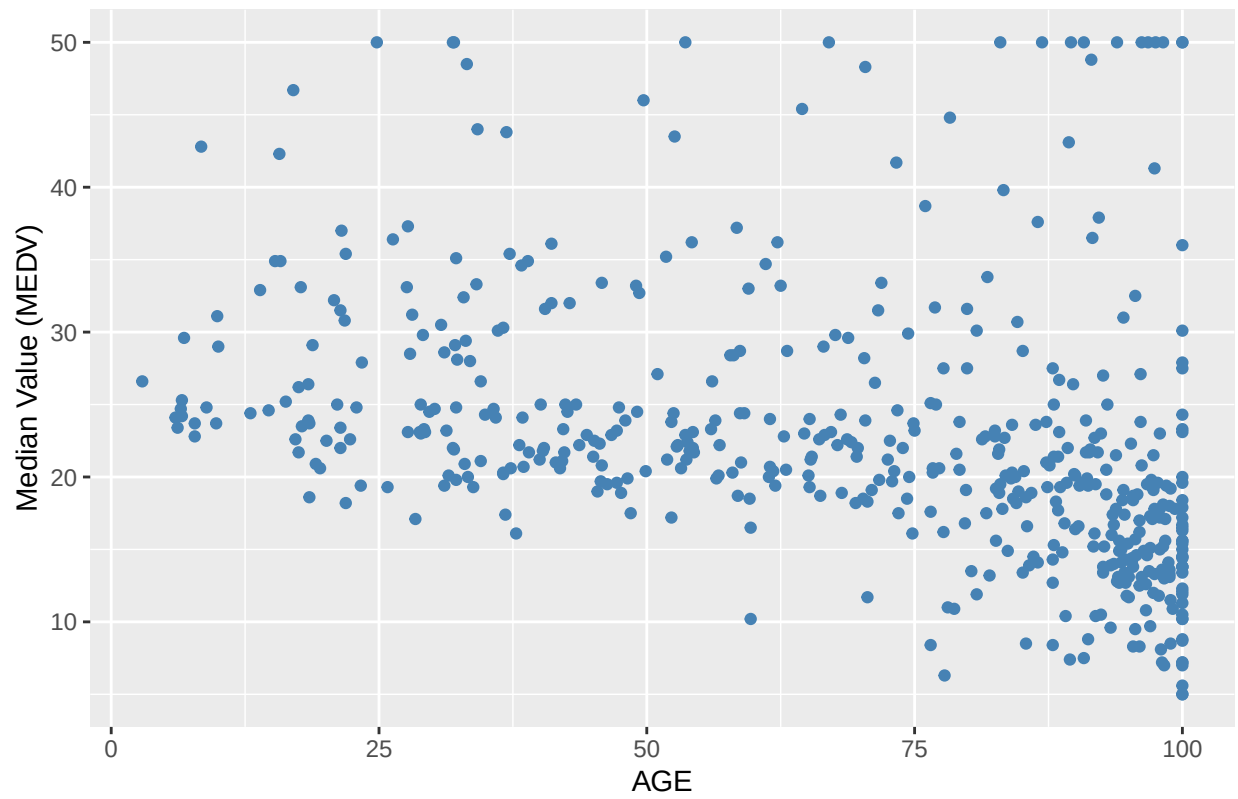


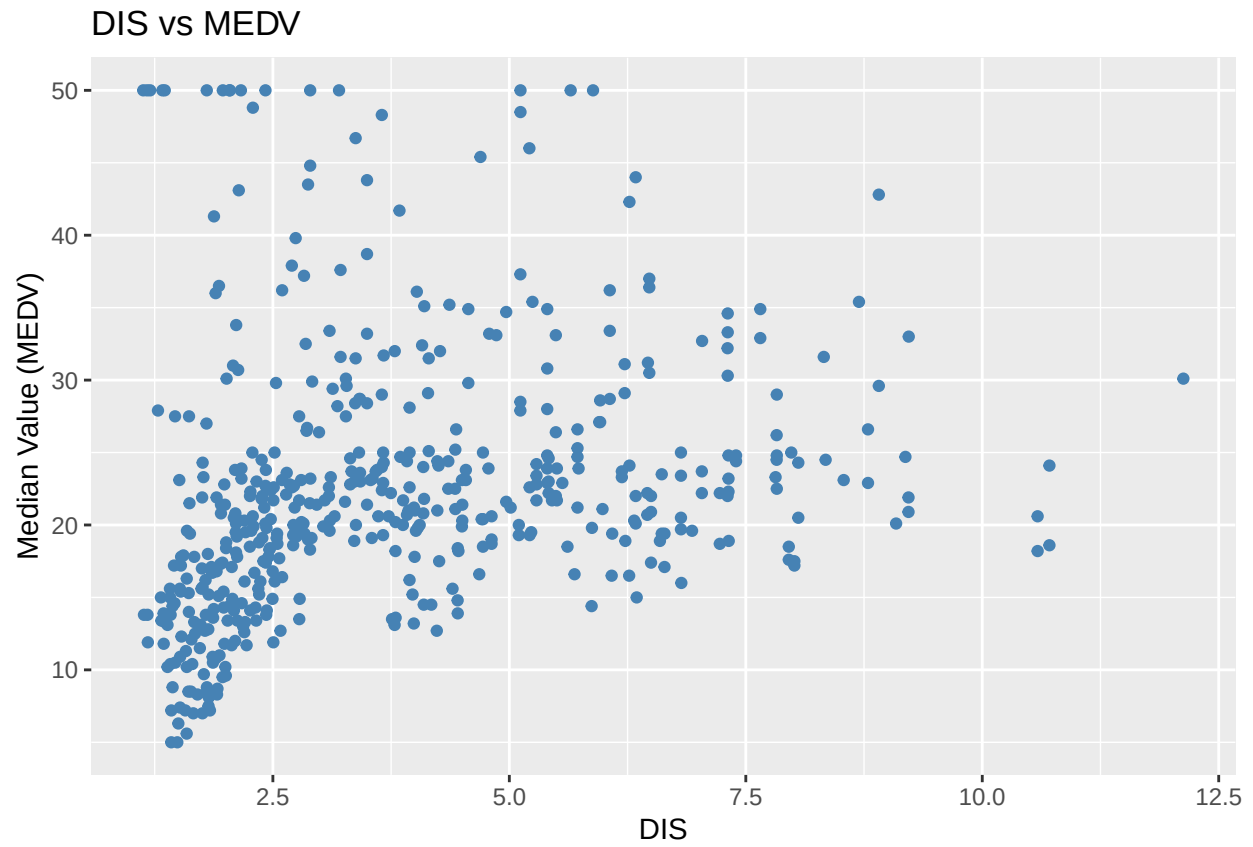


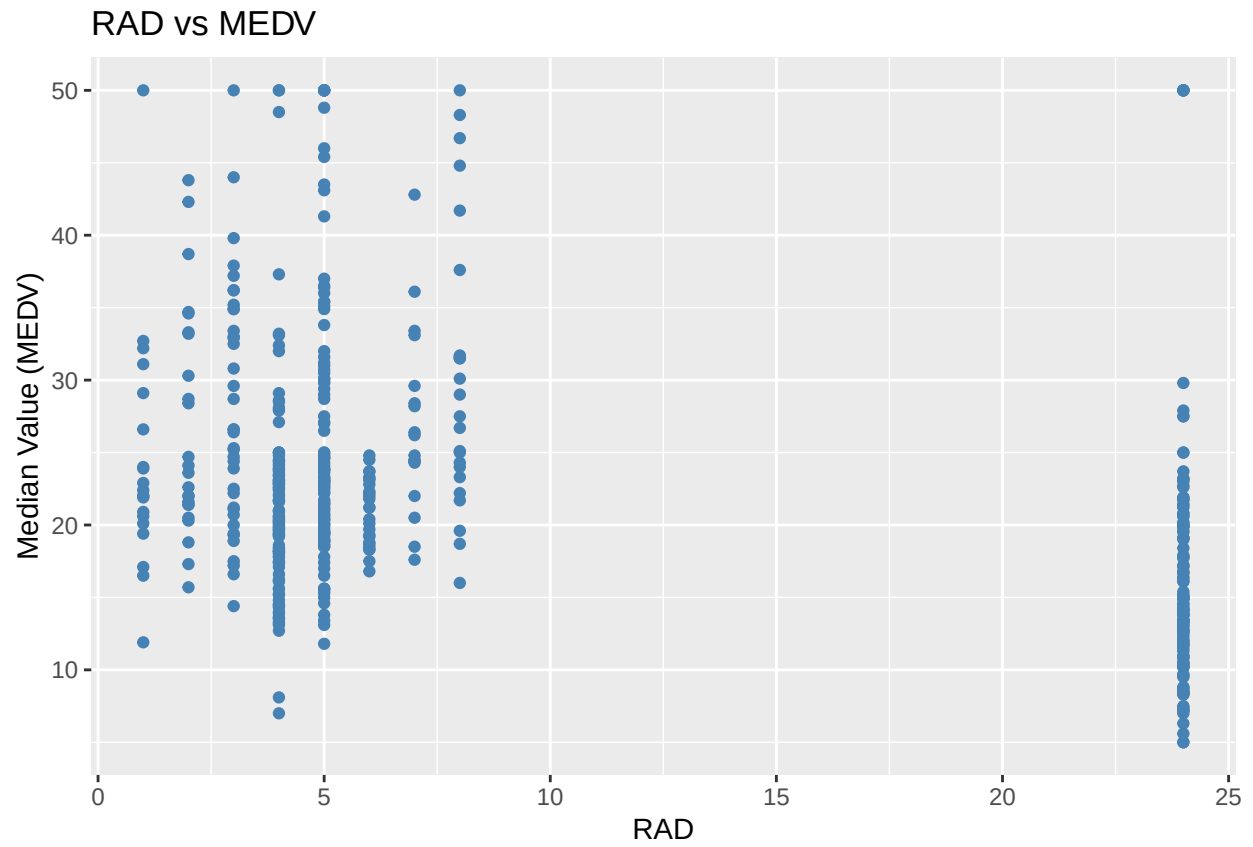


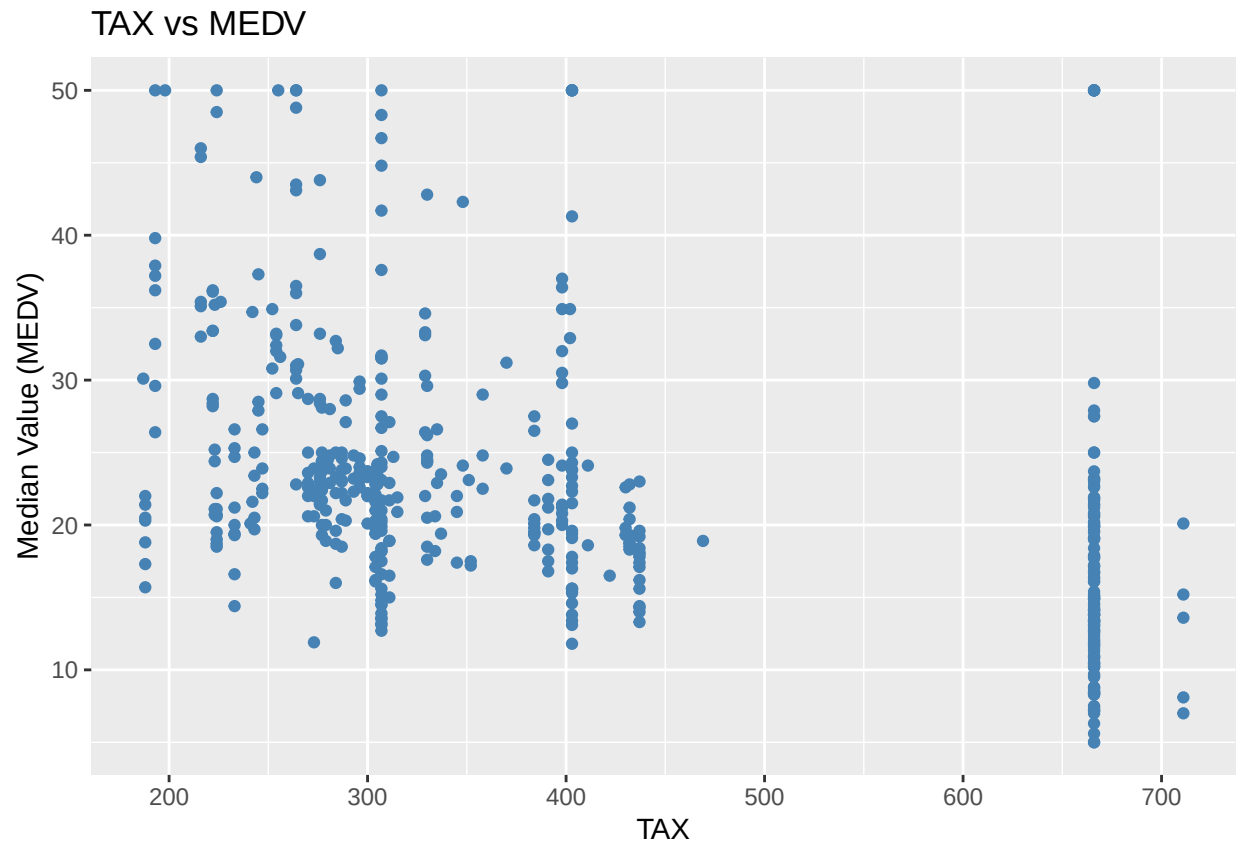


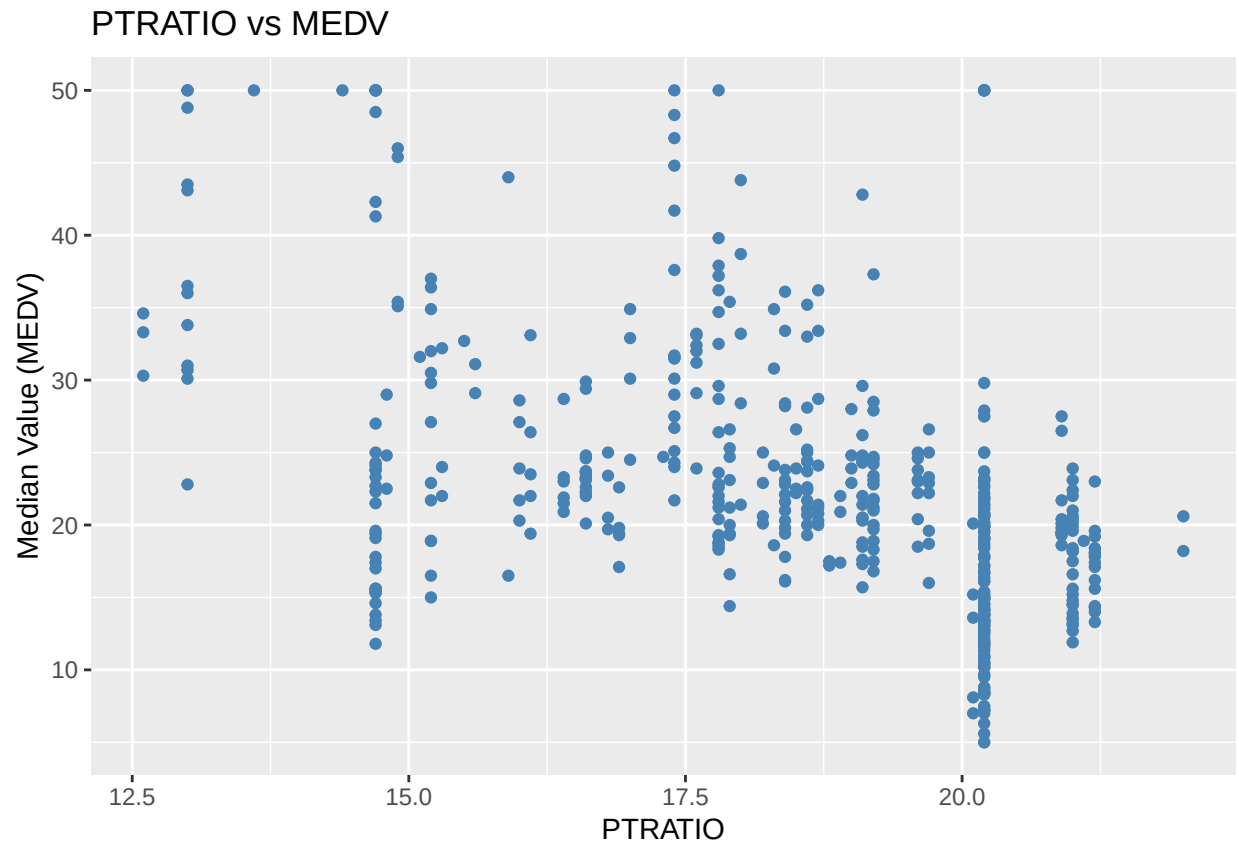
AGE vs MEDV

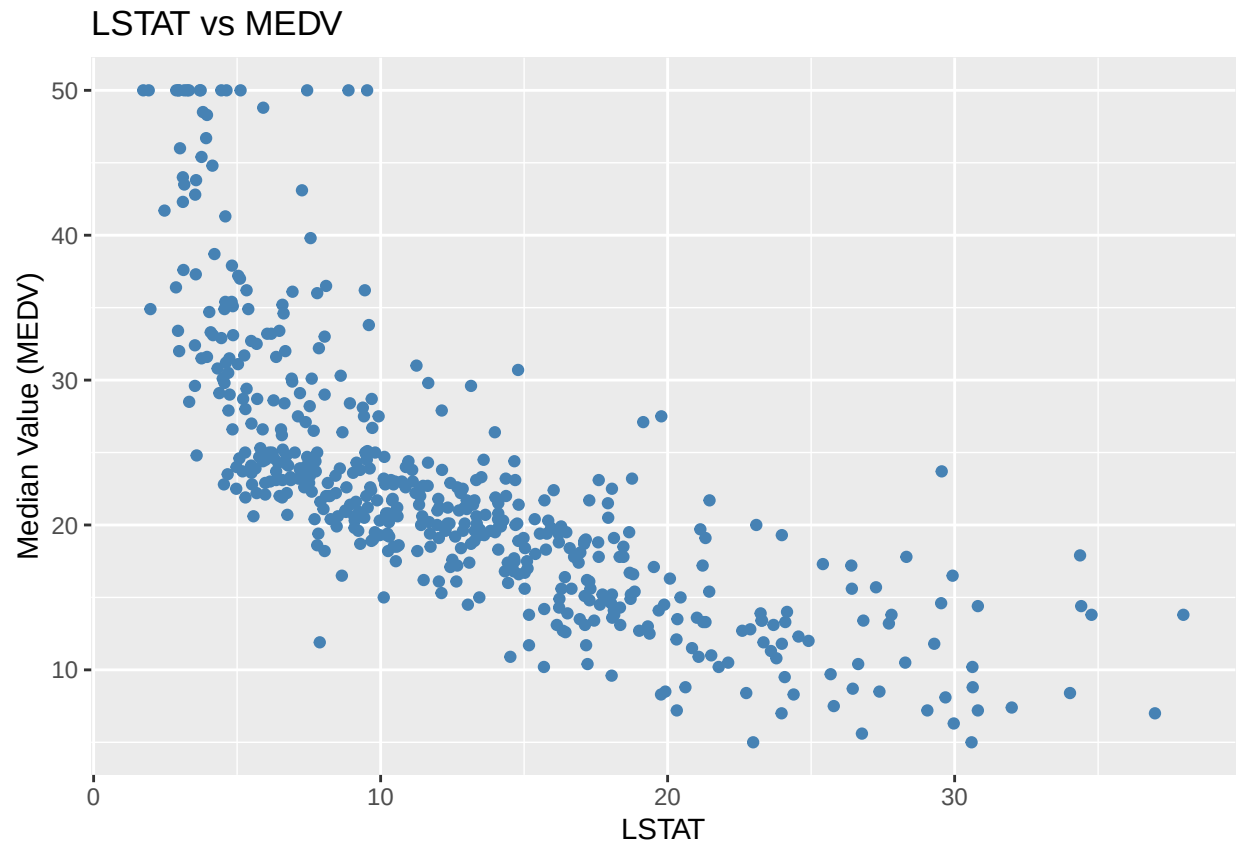






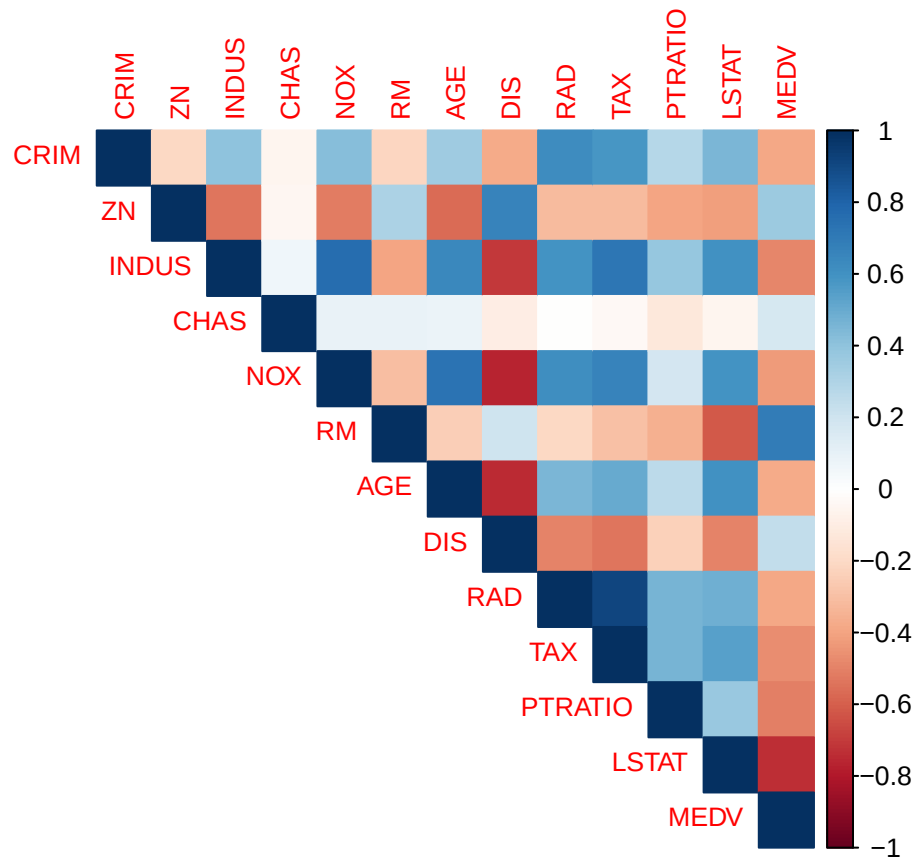






Correlation Heatmap

```
corr_matrix <- cor(data)
corrplot(corr_matrix, method = "color", type = "upper", tl.cex = 0.8, number.cex = 0.7)
```



Composite Plot

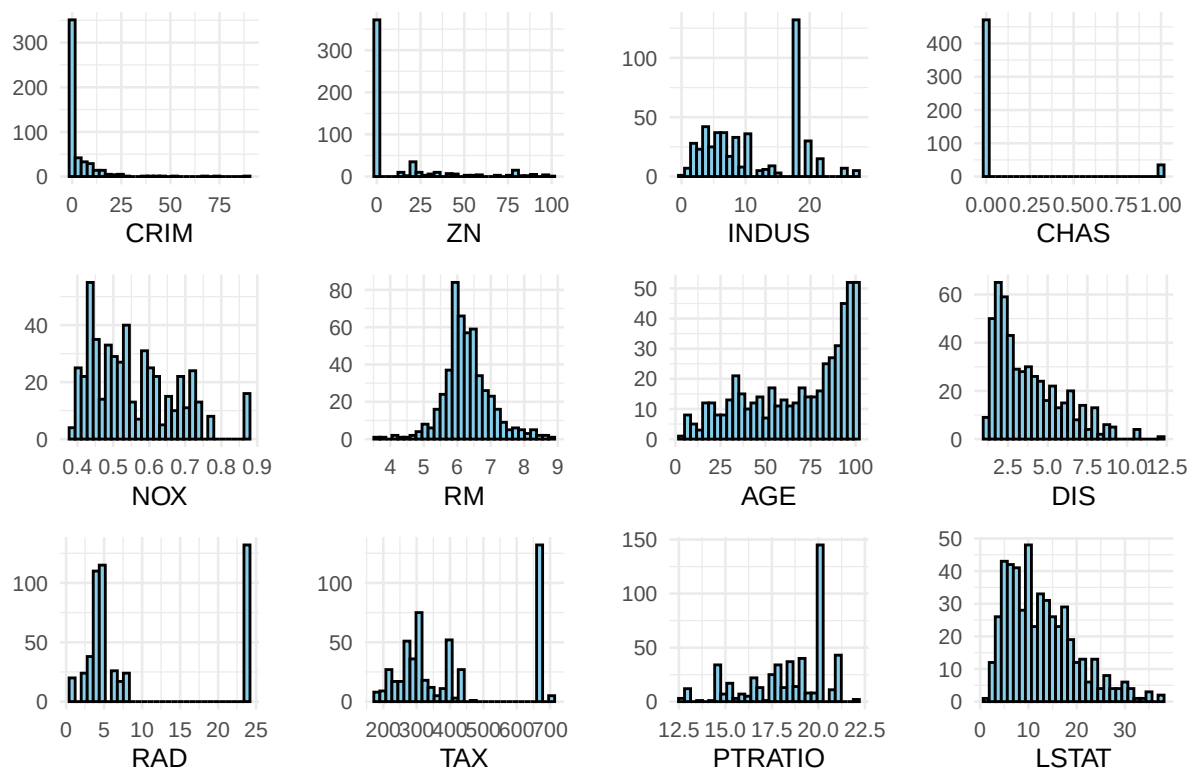
Part 1: Composite Histogram Plot

```
# Create individual histograms
hist_plots <- lapply(predictor_vars, function(var) {
  ggplot(data, aes_string(x = var)) +
    geom_histogram(fill = "skyblue", color = "black", bins = 30) +
    #labs(title = paste("Histogram of", var), x = var, y = "") +
    labs(x = var, y = "") +
    theme_minimal(base_size = 10)
})

# Combine using patchwork
combined_hist <- wrap_plots(hist_plots, ncol = 4) +
  plot_annotation(title = "Visualization: Distribution of Predictors")

combined_hist
```

Visualization: Distribution of Predictors



Part 2: Composite Scatter/Box Plot (Predictors vs MEDV)

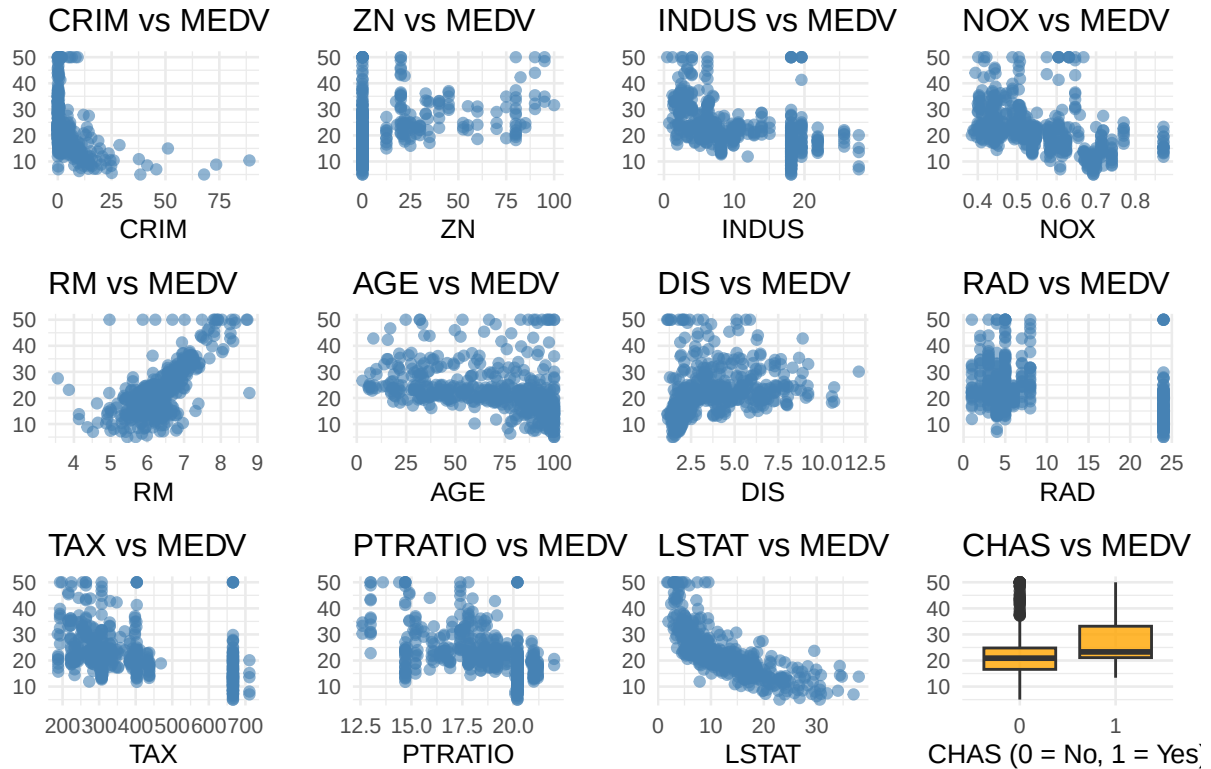
```
# Create scatter plots
scatter_plots <- lapply(numeric_vars, function(var) {
  ggplot(data, aes_string(x = var, y = "MEDV")) +
    geom_point(color = "steelblue", alpha = 0.6) +
    labs(title = paste(var, "vs MEDV"), x = var, y = "") +
    theme_minimal(base_size = 10)
})

# Add CHAS boxplot
chas_box <- ggplot(data, aes(x = factor(CHAS), y = MEDV)) +
  geom_boxplot(fill = "orange", alpha = 0.8) +
  labs(title = "CHAS vs MEDV", x = "CHAS (0 = No, 1 = Yes)", y = "") +
  theme_minimal(base_size = 10)

# Combine all
combined_scatter <- wrap_plots(c(scatter_plots, list(chas_box)), ncol = 4) +
  plot_annotation(title = "Visualization: Predictor Relationships with MEDV")

combined_scatter
```

Visualization: Predictor Relationships with MEDV



Save a high-resolution image

```
ggsave("../output/vis_histograms.png", combined_hist, width = 20, height = 12)
ggsave("../output/vis_scatter.png", combined_scatter, width = 20, height = 12)
```

Summary

This report provides:

- Full visualization of all 12 predictors
- Clear relationships between each variable and the target (MEDV)
- A correlation map to detect multicollinearity